A Unified Framework for Multiscale, Spatial, and Cross-Domain Prediction

Mathematical equations, components, rules, failure diagnosis, repairs, and scenario design

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Research lineage and source basis

This generalized guide is derived from the author's n/s and cross-scale research program archived on Zenodo under DOI 10.5281/zenodo.23111783, together with the subsequent Multi-Lens WPM experiments summarized in the framework paper. The Zenodo record is cited here as the research lineage from which the generalized system-coordinate, scale/location/context, cross-scale, and target-conditioned forecasting ideas were developed.

$$\text{n/s cross-scale program} \rightarrow \text{Multi-Lens WPM experiments} \rightarrow \text{generalized framework}$$

0. What Other Frameworks Can Predict and What This Framework Can Predict

The framework is presented alongside established approaches in scalar forecasting, spatial-field forecasting, reachable-set prediction, multi-resolution modeling, domain adaptation, and calibration. The comparison below focuses on mathematical capability and variable meaning rather than claims of priority.

0.1 Predictions that established methods can also make

Prediction object	Established neighboring literature	Framework relation
Scalar / point forecast	time-series forecasting, regression	Directly supported; not distinctive by itself.
Probability / distribution	probabilistic forecasting, Bayesian prediction	Directly supported; not distinctive by itself.
Future spatial field u(x,t+P)	neural operators, PDE forecasting	Supported as one object-valued target.
Future spatial region / reachable set	reachability, occupancy, set estimation	Supported; region prediction itself is established prior art.
Prediction interval / confidence region	conformal and probabilistic prediction	Supported; uncertainty regions are not unique to the framework.
Multi-scale temporal forecast	multi-resolution transformers, adaptive lookback	Supported; multi-scale modeling alone is established.
Cross-domain latent representation	domain adaptation / invariant representation	Related to N_S, but N_S is physically interpreted and broader.

Therefore the framework's claim is not that these output types are unavailable elsewhere. It is that they can be expressed inside one target-conditioned mathematical architecture and can coexist with predictions over representation space itself.

0.2 Additional prediction spaces represented explicitly in this framework

$$\widehat{\mathcal{Y}}_T(t,P) \text{ is object-valued, not necessarily scalar}$$

$$\widehat{\mathcal{Y}}_T \in \mathbb{R} \cup \mathcal{P}(\mathbb{R}) \cup \mathcal{F}(X) \cup 2^X$$

Here R denotes scalar outputs, P(R) probability/distribution-valued outputs, F(X) spatial or functional fields, and 2^X set/region-valued outputs. The important extension is that X need not be ordinary physical space.

$$\widehat{\mathcal{R}}_T(t,P) \subset X_{\text{spatial}} \times S_{\text{scale}} \times S_{\text{dynamical}} \times W \times m \times C$$

The framework can predict a consequential region in a constructed system-resolution manifold: a future region can simultaneously have spatial location, physical scale, dynamical coordinate, historical extent W, and internal resolution m. This is different from merely predicting where a physical trajectory will be located.

$$\mathcal{M}_T = \mathcal{M}_T(W, m, S_{\text{scale}}, S_{\text{spatial}}, S_{\text{dynamical}} \mid P, T, C)$$

$$B_k \subset W \times m \times S_{\text{scale}} \times S_{\text{spatial}} \times S_{\text{dynamical}}$$

The resolution landscape and its pockets are therefore themselves mathematical prediction objects. A pocket is not merely a confidence interval around a scalar forecast; it is a local basin in target-conditioned representation/utility space whose location, geometry, persistence, and relationships to other basins can be modeled.

0.3 The semantic distinction of the framework variables

Variable	Common neighboring meaning	Meaning in this framework
P	forecast steps / horizon	requested future displacement or extent of the target query; changing P can change which representation is useful
W	lookback-window length	historical extent over which target-relevant geometry is resolved
m	samples / patch size / resolution	support density or internal resolution used to represent structure inside W
S	scale or generic state	system coordinate: scale, spatial location, dynamical clock, geometry, structure, or their combination
C	context/covariates	causal target-conditioned context available at issuance
T	forecast variable/task	explicit target semantics; determines which coordinates and resolutions are consequential
M(W,m)	not usually a first-class object	target-conditioned resolution/utility landscape
B_k	confidence region/basin in other fields	local pocket in representation/utility space, possibly extended through S and physical space

Two papers can therefore use similar symbols or both employ 'multi-scale' processing while solving a different mathematical problem. The comparison must be based on the semantic role of each variable and the space in which the predicted object lives.

0.4 What the broader literature suggests is established versus less commonly combined

Established neighboring capability	Representative literature class	What remains different here
Multi-resolution temporal modeling	MTST; adaptive lookback; adaptive multi-scale decomposition	W and m are axes of a target-conditioned utility landscape, not only architectural patch/window choices.
Domain-invariant representation	DAF; RAINCOAT	N_S is a physically interpretable map that may combine scale, space, clock, geometry and structure.
Physical invariant conditioning	PIANO / neural operators	The framework also conditions temporal-resolution choice and target semantics, not only a field-evolution operator.
Spatial/field prediction	neural operators / PDE surrogates	Output may instead be a region in physical + system + resolution space.
Reachable/probabilistic state sets	reachability / set estimation	Framework pockets can occupy representation/utility space rather than only future physical-state space.
Feature and label shift	domain adaptation	N_T is a physical target-semantic transformation, not merely label-distribution correction.

Published methods can predict fields, regions, sets, multiple temporal scales, and domain-adapted representations. The purpose of this comparison is to show how those capabilities relate to the framework's variables and workflow, not to make a claim about novelty or priority.

0.5 Combined framework chain

$$S_D \xrightarrow{\mathcal{N}_S} S_{\text{common}}$$
$$Y_D \xrightarrow{\mathcal{N}_T} Y_{\text{common}}$$
$$(T, P, S, C) \longrightarrow \mathcal{M}(W, m)$$
$$\mathcal{M} \longrightarrow \{B_1, \ldots, B_K\}$$
$$B_k \subset W \times m \times S \times X$$

$$(\mathcal{M}, \mathcal{B}) \longrightarrow M^{\star} \in \{\text{Fixed}, \text{Multi}, \text{Auto}\}$$
$$M^{\star} \longrightarrow R_D \longrightarrow \text{calibration/adaptation only if required}$$

The proposed distinction is the ordered composition: normalize the physical system coordinate; normalize the physical meaning of the target; construct a causal target-conditioned resolution landscape; preserve multiple pockets when useful; select or mix Fixed/Multi/Auto; test compatibility; and use calibration or target adaptation only after representation-level remedies have been exhausted.

1. Master Equation

At issuance time t, predict target T at requested horizon P using only causal information.

$$\widetilde{S}_D(t) = \mathcal{N}_{S,D}[S_D(t)]$$
$$\widetilde{Y}_D(t,P) = \mathcal{N}_{T,D}[Y_D(t,P)]$$
$$\pi_{Wm}(t,P,T) = R_{Wm}[P,T,\widetilde{S}_D(t),C(t)]$$
$$\widehat{\mathcal{Y}}_{\text{common}}(t,P) = \sum_{(W,m)} \pi_{Wm} F_T[X_{t-W:t;m}, \widetilde{S}_D(t), C(t), P]$$

N_S makes system coordinates comparable. N_T makes target semantics comparable. W controls historical extent; m controls resolution inside W; C is causal context.

The scalar notation below is shorthand. In the generalized form, replace Y_hat by the object-valued Y_hat_T described in Section 0; the same machinery can return a scalar, distribution, spatial field, physical region, or region in the joint system-resolution space.

2. System Coordinate S

$$S = \{S_{\text{scale}}, S_{\text{spatial}}, S_{\text{dynamical}}, S_{\text{structural}}, \dots\}$$
$$S_T^{\star} = G_S(T, P, C, S)$$

S is not merely a scalar scale variable. It is the available physical coordinate space. Only target-relevant components need to be used.

Component	Examples	Purpose
S_scale	system size, U/J, g/wc, lambda/gamma0	nondimensional physical magnitude
S_spatial	relative position, geometry, overlap	spatial invariance
S_dynamical	intrinsic clock, memory time, coupling time	comparable horizon/time
S_structural	parity, purity, IPR, entropy, covariance	state/regime structure

3. Dual Normalization

$$\mathcal{N}_S : S_D \rightarrow S_{\text{common}}$$
$$\mathcal{N}_T : Y_D \rightarrow Y_{\text{common}}$$
$$\text{coordinate difference} \neq \text{domain incompatibility}$$
$$\text{equal numerical threshold} \neq \text{equal physical target}$$

N_S and N_T are separate first-class operations. QKR and non-Markovian experiments in the source paper independently showed that correcting S alone was insufficient when the target itself changed meaning across regimes.

4. Fixed, Multi, and Auto

Fixed

$$\widehat{Y}_F = F_T[X_{t-W_0:t;m_0}, \widetilde{S}, C, P]$$

Fixed is the stable reference and fallback.

Multi

$$\hat{Y}_M = \sum_k \frac{1}{K} F_k[R_k(t)]$$

Multi retains several resolutions and is useful when the correct resolution is uncertain.

Auto

$$\pi_k = R_{Wm}(P, T, \tilde{S}, C)$$
$$\hat{Y}_A = \sum_k \pi_k F_k[R_k(t)]$$

Auto conditionally concentrates or redistributes resolution weight.

Continuous operating mode

$$\mathcal{M} = \{\text{Fixed}, \text{Multi}, \text{Auto}\}$$
$$\hat{L}_j = F_{U,j}(P, \tilde{S}, C, \mathcal{M}_{Wm})$$
$$\alpha_j = \frac{e^{-\lambda \hat{L}_j}}{\sum_k e^{-\lambda \hat{L}_k}}$$
$$\hat{Y} = \sum_j \alpha_j \hat{Y}_j$$

The modes can be treated as a continuous utility state. Do not assume one mode is globally best.

5. Resolution Landscape and Pockets

$$\mathcal{M}_t(W, m \mid P, T) = \text{current causal resolution landscape}$$
$$\mathcal{B}_t = \text{Basins}[\mathcal{M}_t] = \{B_{t,1}, \ldots, B_{t,K_t}\}$$
$$S_{\text{pocket}} = \{\text{count, centroid, width, prominence,}$$
$$\text{curvature, persistence, spacing}, \ldots\}$$

The full landscape preserves all candidate resolution structure. Pockets preserve local low-utility basins. A single optimum is only a compressed special case.

Object	Preserves	Use
Full landscape	all W,m cells	prediction, routing, uncertainty
Pockets	local basins	recognition, reliability, mode selection
Single optimum	one W,m cell	simple stable fallback

5A. Spatial, Field, and Region-Valued Prediction

$$\hat{\mathcal{Y}}_{\text{field}}(x, t, P) = F_T[P, \mathcal{N}_S(S_D), C, R_{Wm}](x)$$
$$\hat{\mathcal{R}}_T(t, P) = \{z \in Z : q_T(z \mid t, P, S, C) \geq \tau\}$$
$$Z = X_{\text{spatial}} \times S_{\text{scale}} \times S_{\text{dynamical}} \times W \times m$$

For field prediction, the framework returns a function over physical coordinates. For region prediction, it returns a subset of a chosen state/manifold Z. Z can be ordinary physical space, but it can also be a product space containing spatial position, scale, dynamical coordinates, and representation coordinates.

$$\text{Pocket}_k = \text{connected/local basin of } \mathcal{M}_T \text{ over } Z$$

This is the key distinction between a physical reachable set and a resolution pocket. A reachable set answers which future physical states are possible. A resolution pocket answers where a target-relevant low-utility or consequential basin exists in the representation/system manifold. The two can be coupled when physical space is included in Z.

Examples of region-valued use

Scenario	Predicted object	Natural coordinates
Spatial driven quantum system	future consequential spatial region	x, relative geometry, S, P
Cross-scale physical system	future compatible scale region	S_scale, P, W, m
Resolution selection	future/current useful representation pockets	W, m, T, P, C
Spatiotemporal field	future field or exceedance region	x, t+P, S
Risk / instability	region of state space with elevated event probability	state, S, P

6. Compatibility and Repair Hierarchy

$$R_D = R_D(T, P, S_A \to S_B, C)$$

Order	Question	Repair
1	Does the target mean the same physical event?	Apply N_T.
2	Are system coordinates comparable?	Apply N_S.
3	Is one historical resolution sufficient?	Use Multi/Auto.
4	Does one optimum hide useful structure?	Retain landscape/pockets.
5	Does routing transfer prospectively?	Use compatibility gate / fallback.
6	Does ranking transfer but probability scale fail?	Light calibration.
7	Does structural discrimination fail?	Target-domain adaptation.
8	Is adaptation stable prospectively?	If no: specialized model, fallback, or abstain.

$$\mathcal{N}_S \to \mathcal{N}_T \to (W, m) \ \text{landscape} \to \text{pockets/mode} \to \text{compatibility} \to \text{calibration/adaptation}$$

7. Diagnostic Pattern: Correct S, Wrong Target

Quantum kicked rotor

$$D_{\text{ref}}(K) \approx \frac{K^2}{2}$$

$$R_p(t, h) = \frac{\text{Var}[p(t+h)] - \text{Var}[p(t)]}{(K^2/2)h}$$

$$E_{\text{transport}} = 1\left[\max_{1 \le h \le P} R_p(t, h) \ge R^\star\right]$$

Absolute momentum width produced a non-comparable event across K. Target normalization converted it into transport relative to the natural kick-induced diffusion scale.

Non-Markovian open dynamics

$$\gamma(t) = -\frac{d}{dt}\log P_e(t)$$

$$\mathcal{B}(t, P) = \int_t^{t+P\Delta t} \max[-\gamma(\tau), 0]\, d\tau$$

$$P_{\text{memory}} = \lambda P \Delta t$$

$$E_{\text{NM}} = 1[\mathcal{B}(t, P) \ge \mathcal{B}^\star]$$

Absolute population revival was replaced by future information backflow. This aligned the target with the physical meaning of non-Markovianity.

8. Diagnostic Pattern: Spatial or Size Coordinates Differ

$$\mathcal{N}_i : S_i \to S_{\text{common}}$$

$$x_{\text{rel}} = \frac{x_{\text{state}} - x_{\text{structure}}}{L}$$

$$\tilde{\rho}(\xi) = \mathrm{Interp}[\rho_i(x/L_i)]$$
$$\tilde{\epsilon}(\xi) = \mathrm{Interp}\left[\frac{\epsilon_i - \bar{\epsilon}_i}{\sigma_{\epsilon_i}}\right]$$

Use relative geometry and nondimensional coordinates when native locations or lattice sizes differ. The normalizer should preserve the physical relationship, not merely standardize values.

9. Diagnostic Pattern: One Resolution Fails

The Rabi crossing in the source paper withheld the complete intermediate coupling interval. Fixed common-S remained weak, while Multi strongly recovered zero-shot prediction.

$$D_{\text{source}} = [0.02, 0.08] \cup [0.28, 0.48]$$
$$D_{\text{target}} = [0.10, 0.22]$$

Interpretation: unseen transition regimes can be representationally compatible while the correct temporal resolution remains uncertain. In that case distributed Multi can be safer than concentrated Auto.

The non-Markovian memory crossing instead favored Auto. Therefore mode choice is conditional, not a universal rule.

10. Diagnostic Pattern: Ranking Improves, Calibration Worsens

$$\mathrm{AUROC}\uparrow,\ \mathrm{Brier}\uparrow \Rightarrow \text{structural ranking transfers; probability scale does not}$$
$$z = \log\frac{p}{1-p}$$
$$p_{\text{cal}} = \sigma(a\,z + b)$$

Freeze the structural predictor and calibrate only the probability map when monotone calibration repairs Brier while preserving ranking.

$$\text{Adapt only if repeated prospective draws stably beat the reference}$$

Small target samples can be rejected even when a median result looks favorable.

11. Diagnostic Pattern: Mode Encoding Is Wrong

$$\text{Bad: } U < U_0 \to \text{prototype}_A,\ U \geq U_0 \to \text{prototype}_B$$
$$\text{Better: preserve } U/J \text{ continuously in } S$$
$$\text{Bad: } M^\star = \arg\min_j L_j \text{ as the only target}$$
$$\text{Better: learn continuous } \hat{L}_j \text{ and apply softmin}$$

Do not discretize a continuous physical coordinate unless physics supplies a justified boundary. Do not discard utility magnitude when the difference between a weak preference and a decisive preference matters.

12. Rules for Different Scenarios

Scenario	Recommended construction	Primary check
Known regime	Fixed or validated Auto	Does routing improve Brier and ranking?
Unseen parameter interval	N_S + Multi/Auto	Does common representation transfer?
Spatial crossing	relative spatial N_S	Are absolute coordinates hiding invariance?
Different size	finite-size/nondimensional N_S	Do scaled observables collapse?
Different clock	dimensionless time/horizon	Is P physically comparable?
Different target scale	N_T	Does the event have the same physical meaning?
Cross-family	N_S + N_T + common observables	Does transfer survive without target fitting?
Ranking good, Brier bad	freeze structure + calibrate	Does monotone calibration beat reference?
Landscape multimodal	full landscape + pockets	Does one optimum hide useful basins?
Routing unstable	Fixed/Multi fallback	Does compatibility gate prevent harm?

13. How to Construct a New Domain

Step 1 - Define the target mathematically.

$$Y_D(t,P) = \text{target measured over the interval } (t, t+P]$$

Step 2 - Test semantic invariance. If the same threshold has different physical meaning across regimes, define N_T.

Step 3 - Enumerate system coordinates.

$$S_D = \{\text{scale, space, clock, geometry, structure}\}$$

Step 4 - Define N_S using physics before statistical convenience.

Step 5 - Define candidate resolution cells.

$$\Omega_R = \{(W_1, m_1), \ldots, (W_K, m_K)\}$$

Step 6 - Fit Fixed, Multi, and Auto references on source-only development data.

Step 7 - Build the current causal landscape and pockets only if needed.

Step 8 - Evaluate on disjoint locked regimes/domains. Never tune the normalizer or router from locked outcomes.

Step 9 - Diagnose failure by metric signature and apply the smallest repair.

Step 10 - Re-run prospectively after every repair.

14. Metric Signatures and What They Mean

Observed signature	Likely interpretation	Next action
Brier down, AUROC up	clean improvement	retain representation
Brier up, AUROC up	ranking gain, calibration mismatch	calibrate or gate
Brier down, AUROC flat	probability-scale gain	retain if stable
Both worse	representation/routing incompatibility	fallback; inspect N_S/N_T
Extreme prevalence shift	target semantics mismatch	inspect N_T
One subdomain fails	local compatibility failure	subdomain gate/normalizer
Auto worse, Multi better	over-concentration	retain distributed resolution
Fixed best	history/routing adds no prospective value	use Fixed

15. Mathematical Examples from the Research Program

Example	Problem	Repair	Result stated in source paper
Quantum Rabi crossing	unseen intermediate coupling; Fixed weak	common S + Multi W,m	47.91% Brier reduction vs Native Fixed
Non-Markovian memory crossing	withheld memory interval	N_T backflow + memory S + Auto	58.64% Brier reduction
QKR crossing	absolute-width target not comparable	diffusion-normalized N_T + W,m	Auto 2.99% Brier reduction
BH -> hard-core/spinless	different lattice/statistics	common spatial/system N_S	Auto 19.15% Brier reduction
Bose-Hubbard calibration	ranking transferred, probability scale did not	light target calibration	40-trajectory calibration 10.00% below Fixed
Floquet spatial crossing	different drive location	relative spatial normalization	near-invariant zero-shot transfer

16. Minimal Deployment Form

A practical implementation does not need every layer. Start with the smallest equation that survives prospective validation.

$$\hat{Y} = F_T[P, \mathcal{N}_S(S_D), C, R_{W,m}]$$

Add N_T only when target semantics differ. Add Multi/Auto only when Fixed is insufficient. Add pockets only when landscape multimodality matters. Add calibration only when structural ranking transfers but probabilities do not. Add target adaptation only after representation-level remedies are exhausted.

17. Full Deployment Form

$$\tilde{S} = \mathcal{N}_{S,D}(S_D)$$

$$\widetilde{Y} = \mathcal{N}_{T,D}(Y_D)$$
$$\mathcal{M}_{Wm} = \text{Landscape}[X_{\le t}, P, T, \widetilde{S}, C]$$
$$\mathcal{B} = \text{Basins}[\mathcal{M}_{Wm}]$$
$$\hat{L}_j = U_j(P, T, \widetilde{S}, C, \mathcal{M}_{Wm}, \mathcal{B})$$
$$\alpha_j = \text{softmin}_j(\lambda \hat{L}_j)$$
$$\pi_{Wm} = R_{Wm}(P, T, \widetilde{S}, C \mid j = \text{Auto})$$
$$\hat{Y}_{\text{common}} = \sum_j \alpha_j \hat{Y}_j$$
$$\text{if incompatible: } \Gamma_{\text{target}}(\hat{Y}_{\text{common}} \mid \widetilde{S}) \text{ or target adaptation}$$

The full form is a hierarchy, not a requirement to activate every component. Complexity should be earned by prospective evidence.

18. Core Rules

Rule 1: Preserve causality. No feature, landscape, pocket, or normalizer may use unresolved target information at issuance.

Rule 2: Normalize physics before fitting adaptation.

Rule 3: Separate system normalization N_S from target normalization N_T.

Rule 4: Do not equate coordinate difference with domain incompatibility.

Rule 5: Do not force Auto; Fixed and Multi are legitimate operating states.

Rule 6: Preserve full landscape or pockets when one optimum destroys useful structure.

Rule 7: Diagnose AUROC and Brier separately.

Rule 8: Prefer calibration over structural retraining when ranking already transfers.

Rule 9: Adapt only after reasonable representation-level repairs are exhausted.

Rule 10: Freeze protocols and use disjoint development, validation, and locked tests.

19. Compact Decision Equation

$$\text{Given } (D, T, P):$$
$$1)\ \widetilde{Y} = \mathcal{N}_{T,D}(Y_D)$$
$$2)\ \widetilde{S} = \mathcal{N}_{S,D}(S_D)$$
$$3)\ M^\star \in \{\text{Fixed}, \text{Multi}, \text{Auto}\} \text{ from prospective utility}$$
$$4)\ M^\star = \text{Auto} \Rightarrow \text{choose } \pi_{Wm}$$
$$5)\ \text{forecast } \hat{Y}_{\text{common}}$$
$$6)\ \text{test compatibility } R_D$$
$$7)\ \text{if ranking good but calibration bad} \Rightarrow \Gamma$$
$$8)\ \text{if structural transfer bad} \Rightarrow \text{target adaptation}$$
$$9)\ \text{if unstable} \Rightarrow \text{fallback/abstain}$$

19A. Relationship to Published Work

When comparing this framework to published methods, compare mathematical roles rather than names. Neural operators may predict spatial fields; reachability may predict state sets; adaptive-lookback methods may select W; multi-resolution models may fuse scales; domain adaptation may learn invariant features. These are important prior-art neighbors. The proposed distinction is not any one capability in isolation.

$$\text{Framework object} = (\text{dual physical normalization})$$
$$+ (\text{target-conditioned } W, m \text{ landscape}) + (\text{pockets})$$

$$+(\text{mode state})+(\text{compatibility/repair hierarchy})$$

The comparison should remain capability-based: describe what neighboring published methods predict or transform, then describe what this framework predicts or transforms using its own variable semantics.

19B. Selected Published Work for Context

The references below are included to strengthen the mathematical context. They illustrate capabilities that overlap with parts of the framework: simultaneous temporal resolutions, adaptive historical extent, domain adaptation, feature/label shift, and physical-invariant operator learning. They are used as neighboring technical references rather than as statements that the framework is the same as, or different in priority from, those methods.

- **Zhang et al. (2024), Multi-resolution Time-Series Transformer for Long-term Forecasting, AISTATS / PMLR 238, pp. 4222-4230.** Uses multiple temporal patch resolutions simultaneously to model patterns at different frequencies.
- **Tong and Yuan (2026), Efficiently Enhancing Long-term Series Forecasting via Adaptive Lookback with Wavelets, AAAI 2026.** Dynamically determines useful lookback length using wavelet-derived multi-scale information.
- **Jin et al. (2022), Domain Adaptation for Time Series Forecasting via Attention Sharing, ICML / PMLR 162.** Learns domain-invariant temporal features so a target forecasting domain can leverage source-domain data.
- **He et al. (2023), Domain Adaptation for Time Series Under Feature and Label Shifts (RAINCOAT), ICML / PMLR 202.** Addresses time/frequency feature shifts and label shifts in time-series domain adaptation.
- **Liu et al. (2024), Deciphering and integrating invariants for neural operator learning with various physical mechanisms (PIANO), National Science Review 11(4), nwad336.** Extracts and conditions neural operators on physical invariants across varying PDE mechanisms.
- **Nguyen, Thy. Zenodo research record, DOI: 10.5281/zenodo.23111783.** Research lineage for the author's n/s cross-scale program from which this generalized framework is derived.

References

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20. Closing Interpretation

The framework is best understood as a controlled sequence of invariance tests. First ask whether systems can share coordinates. Then ask whether targets share meaning. Then ask which historical resolution is needed. Only after those questions are answered should a failure be called domain incompatibility. This ordering prevents target adaptation from being used to compensate for a bad coordinate system, a bad target definition, or an unnecessarily collapsed resolution landscape.